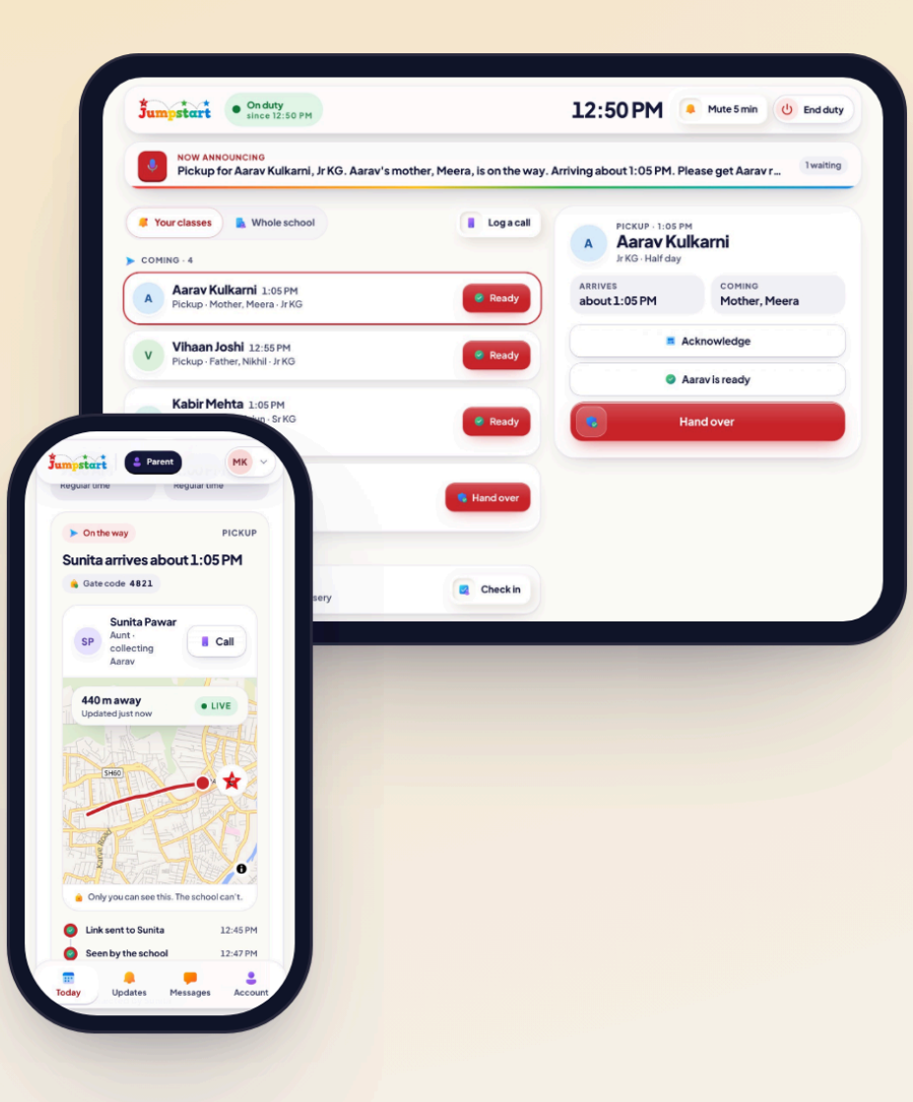
JUMPSTART GO · SYSTEM GUIDE

How Jumpstart Go *works.*

One installable web app for parents, the gate and the office. One Postgres database that enforces every rule. A self-hosted Indian voice at the gate. No servers of our own to patch, and ₹0 a month today.

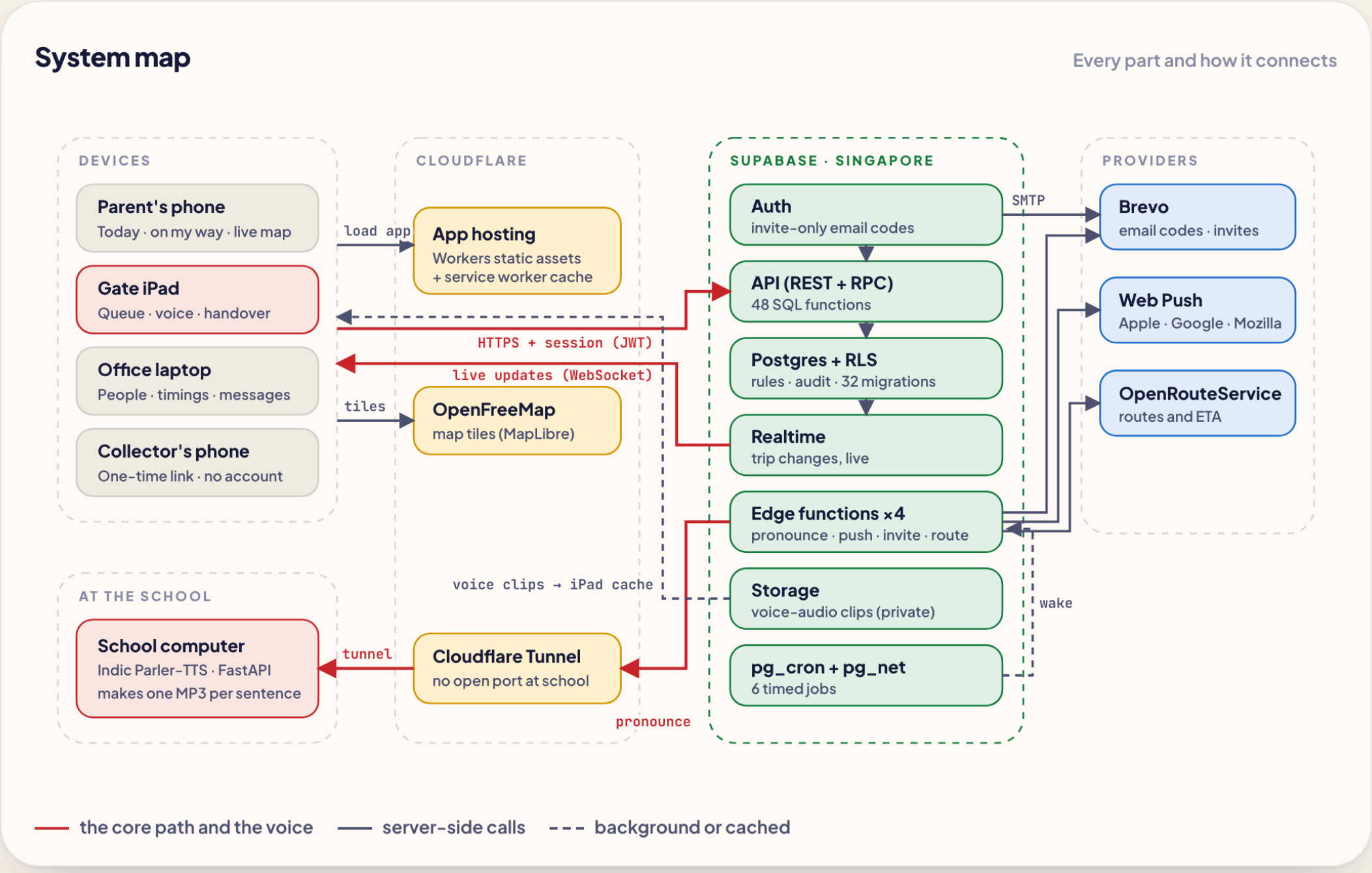


- **1** web app for every role
- **48** database functions guard every change
- **4** edge functions hold the secrets
- **₹0** per month on free tiers today

01 · THE BIG PICTURE

Four layers, *one* source of truth.

Every device runs the same app. It talks only to Supabase, which checks who is asking on every request. Anything that needs a secret, such as sending email, push or asking the voice, runs in a small server-side function.



WHO USES IT

Devices

- **Parent's phone** Today, I'm on my way, live map
- **Gate iPad** Queue, voice announcements, handover
- **Office laptop** People, timings, attendance, messages
- **Collector's phone** One-time link, no account

DELIVERY

Cloudflare

- **Workers static assets** Serves the built app from the edge. No server code.
- **Service worker (PWA)** Installs to the Home Screen, caches the app shell, prompts before updating
- **Tunnel** Reaches the school's voice computer without opening a port

THE CORE · SINGAPORE

Supabase

- **Auth** Invite-only, 6-digit email codes
- **Postgres + row-level security** 32 migrations; every state change is a SQL function
- **Realtime** Pushes trip changes to the gate and parents
- **Edge functions + pg_cron** Email, push, routing, voice; 6 timed jobs
- **Storage** Private voice-clip bucket

OUTSIDE SERVICES

Providers

- **School computer** AI4Bharat Indic Parler-TTS
- **Brevo** Sign-in codes and invitations
- **Web Push** Apple, Google, Mozilla push services
- **OpenFreeMap + OpenRouteService** Map tiles and ETA routes

02 · THE TECHNOLOGY

What each part is *built with.*

Frontend

One React app with role areas for admin, head office, front desk, center leaders and parents. Server rendering isn't needed behind a login.

`React 19` `TypeScript 7` `Vite 8` `React Router 7` `Tailwind CSS 4` `lucide icons`

App on devices

An installable web app (PWA): no App Store. The gate iPad holds a screen wake lock while on duty.

`vite-plugin-pwa` `Workbox` `Web Push` `Wake Lock` `Web Audio`

Database

Postgres is the enforcement point: constraints, row-level security and a SQL function for every change, each writing an audit entry.

`Postgres` `RLS` `pg_cron` `pg_net` `32 migrations`

Backend

No app server. Four Deno edge functions hold the secrets; everything else is SQL called through Supabase's API.

`Supabase` `Edge Functions (Deno)` `supabase-js 2`

Voice

A Python service on the school's computer turns one sentence into an MP3, loaded once, one sentence at a time, behind a shared secret.

`Indic Parler-TTS` `PyTorch` `FastAPI` `Cloudflare Tunnel`

Quality

Pure TypeScript rules in `src/domain` give instant feedback; the database repeats the rules that protect data.

`Vitest · 116` `SQL tests` `pytest · 20` `MapLibre GL 6`

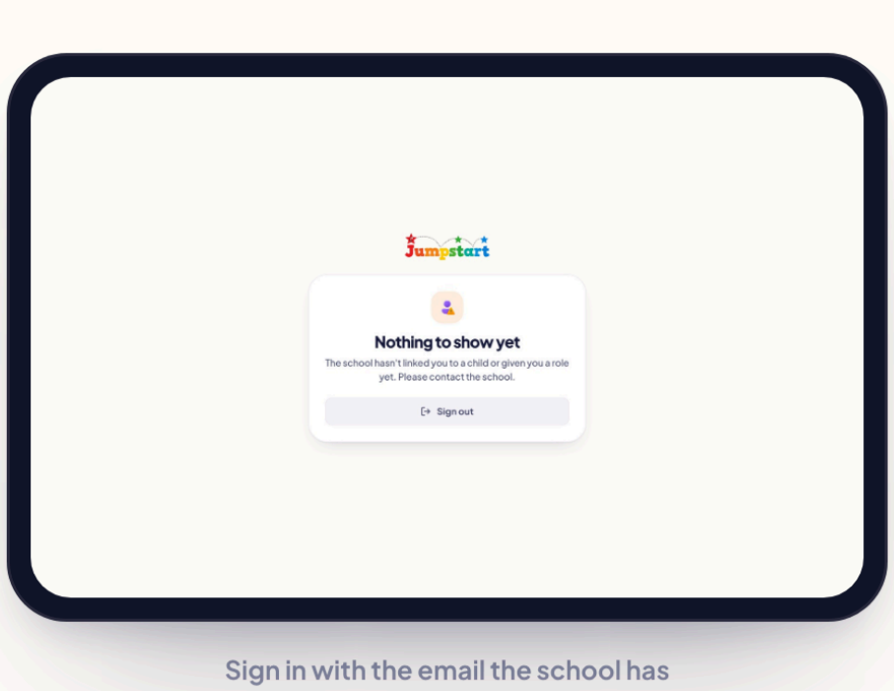
03 · THE FLOWS

What happens when someone *taps.*

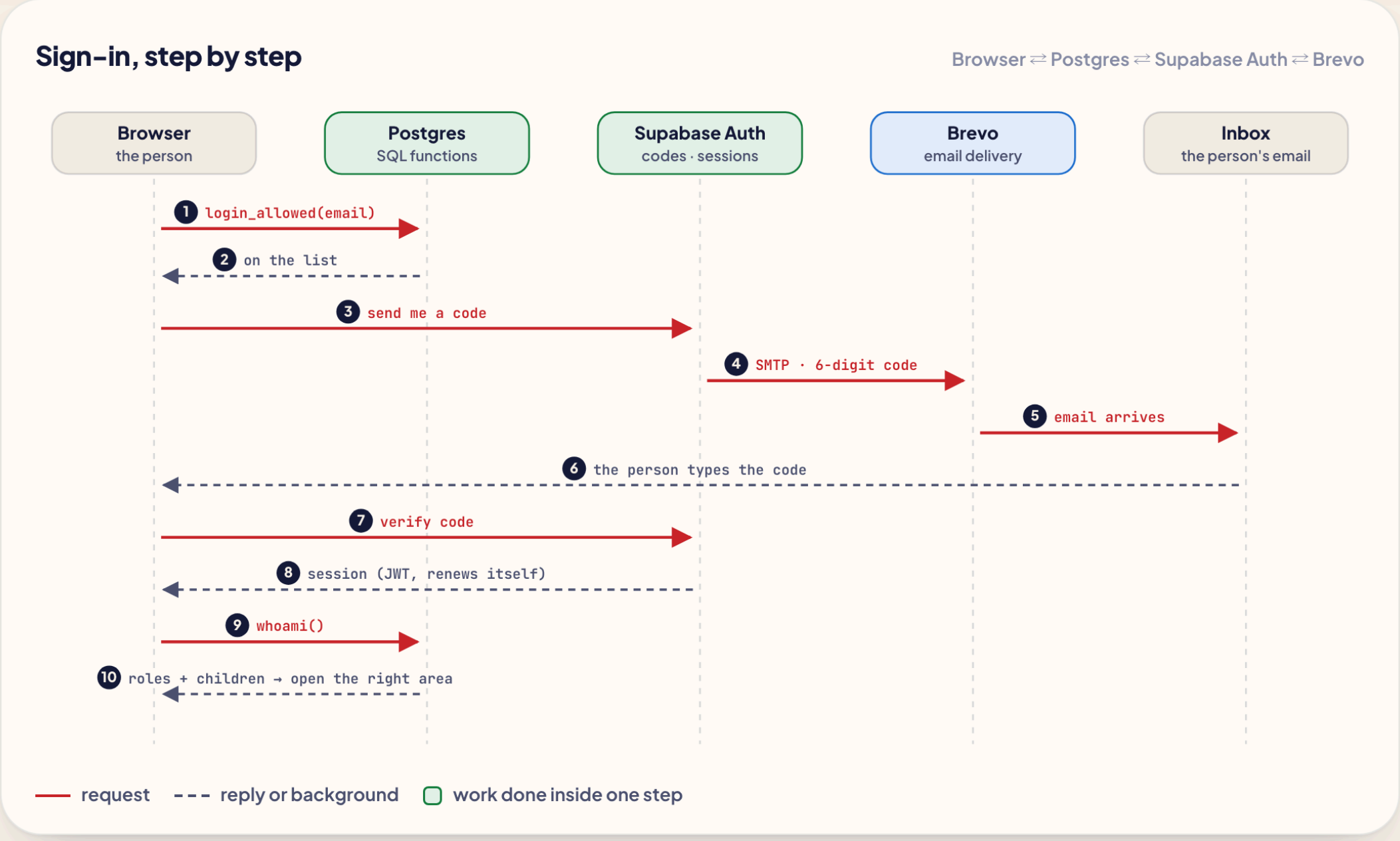
Signing in

Invite-only. There are no passwords to leak and no sign-ups.

1. **Enter the school-given email**
   The app asks `login_allowed` whether the address is on the school's list.
   Database
2. **A 6-digit code is emailed**
   Supabase Auth creates the code; Brevo SMTP delivers it. It works once, for 5 minutes.
   Brevo
3. **Enter the code: a signed session (JWT)**
   Every later request carries it; row-level security reads who you are from it.
   Auth
4. **The app asks `whoami`**
   It returns roles and linked children, and opens the right area. Expired sessions are renewed and retried automatically.
5. **Parents agree to the privacy notice once**
   `accept_consent` records the notice version.



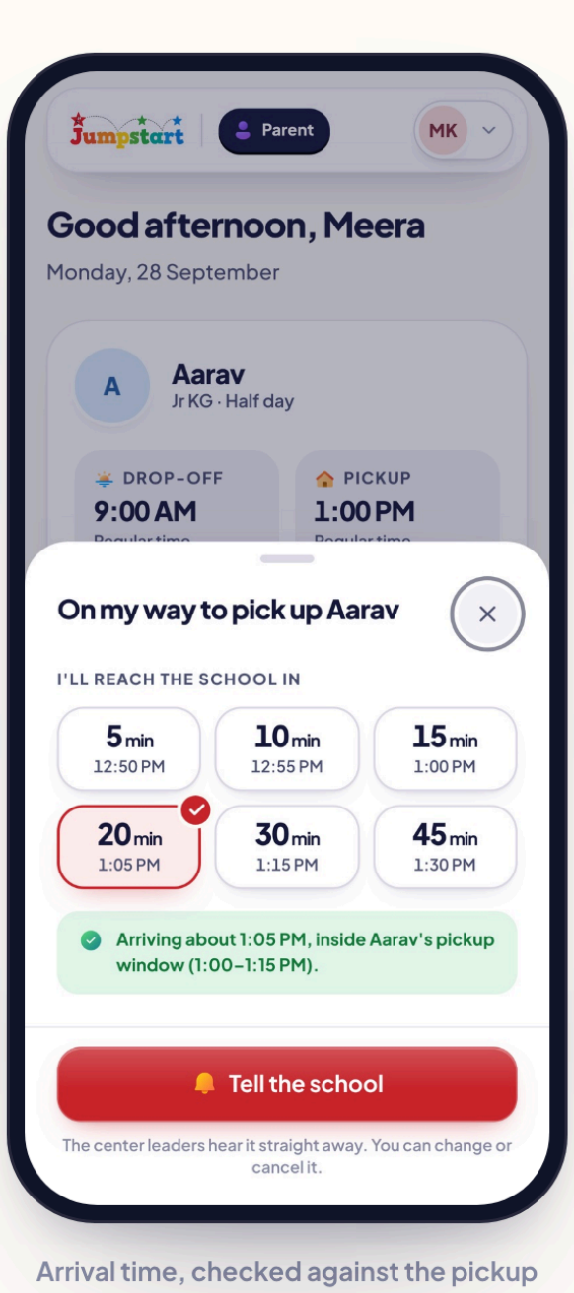
Sign in with the email the school has



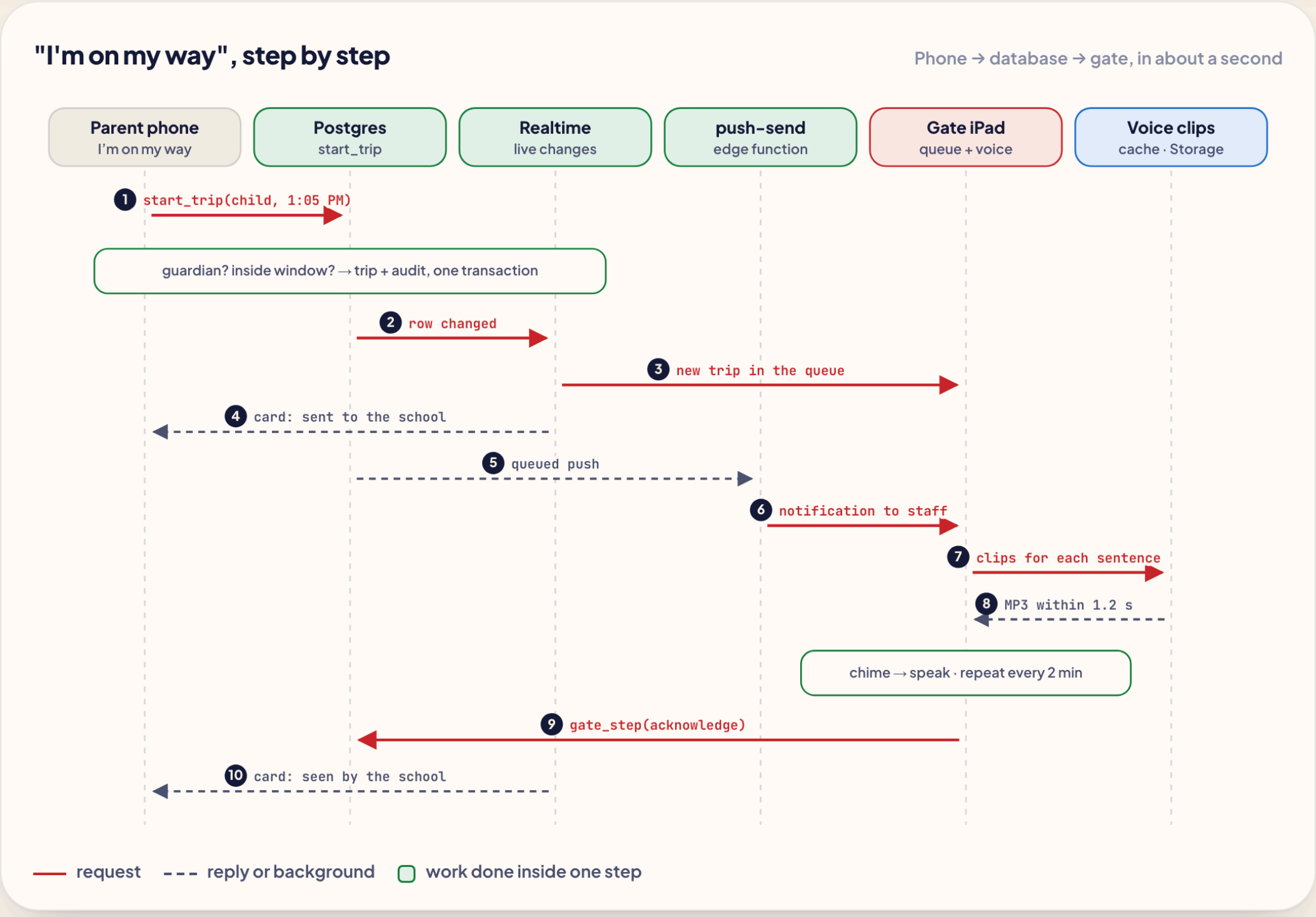
"I'm on my way"

The core pickup flow: from a parent's tap to the gate knowing, in about a second.

1. **The parent picks an arrival time**
   The app shows whether it falls inside the child's pickup window. Outside it, a reason is asked for.
   Parent
2. **`start_trip` runs in Postgres**
   It checks the caller is this child's guardian, validates the time, writes the trip and an audit entry in one transaction.
   Database
3. **Realtime tells every gate iPad**
   The trip appears in the queue; the office board updates.
   Realtime
4. **Push notifications go out**
   `push-send` signs messages with the school's VAPID key for subscribed staff and family.
   Edge function
5. **The gate announces it**
   Chime, then the sentence in the school's own voice. The next section explains how.
   Gate



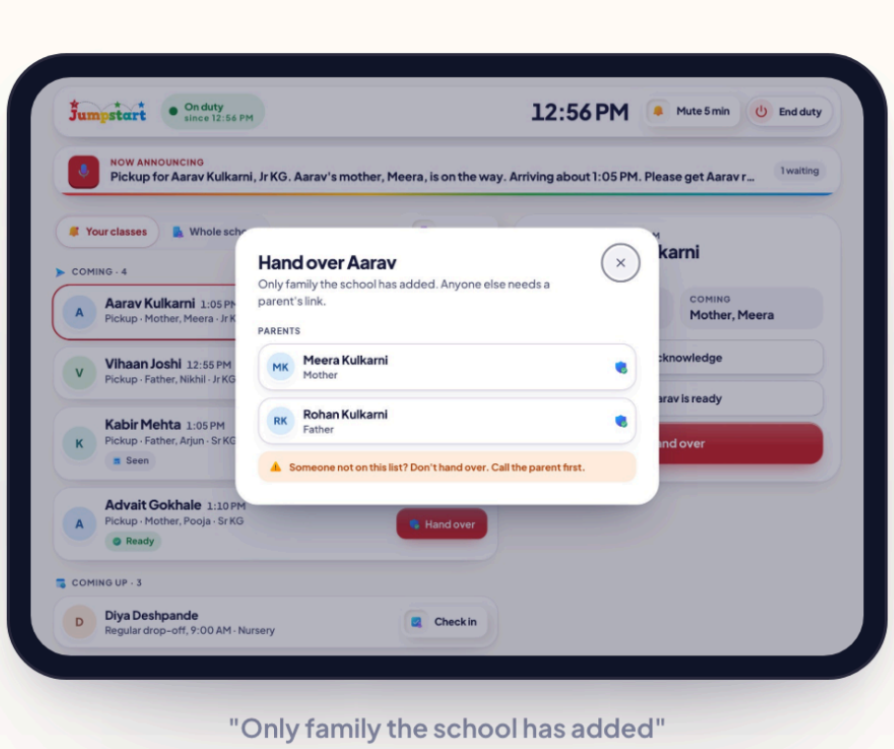
Arrival time, checked against the pickup window


"I'm on my way", step by step
Phone → database → gate, in about a second
Parent phone
I'm on my way
Postgres
start_trip
Realtime
live changes
push-send
edge function
Gate iPad
queue + voice
Voice clips
cache · Storage
1 start_trip(child, 1:05 PM)
guardian? inside window? → trip + audit, one transaction
2 row changed
3 new trip in the queue
4 card: sent to the school
5 queued push
6 notification to staff
7 clips for each sentence
8 MP3 within 1.2 s
chime → speak · repeat every 2 min
9 gate_step(acknowledge)
10 card: seen by the school
request
reply or background
work done inside one step

At the gate: acknowledge, ready, hand over

Center leaders work from the queue. The database records each step and who took it.

1. **Acknowledge**
   `gate_step('acknowledge')` stops the 2-minute repeats on every iPad.
2. **Mark the child ready**
   The parent's card ticks "ready at the gate".
3. **Hand over only to known family**
   Parents and family the school has added are listed. Someone sent with a link must give their code: `gate_handover_code` locks after three wrong tries.
   Safety
4. **Walk-ins and phone calls**
   `gate_walk_in` and `log_call` cover people who arrive without the app.


Jumpstart
12:56 PM
Hand over Aarav
Only family the school has added. Anyone else needs a parent's link.
PARENTS
Meera Kulkarni
Mother
Rohan Kulkarni
Father
Someone not on this list? Don't hand over. Call the parent first.
Diya Deshpande
Check in

"Only family the school has added"

04 · THE VOICE ANNOUNCEMENT

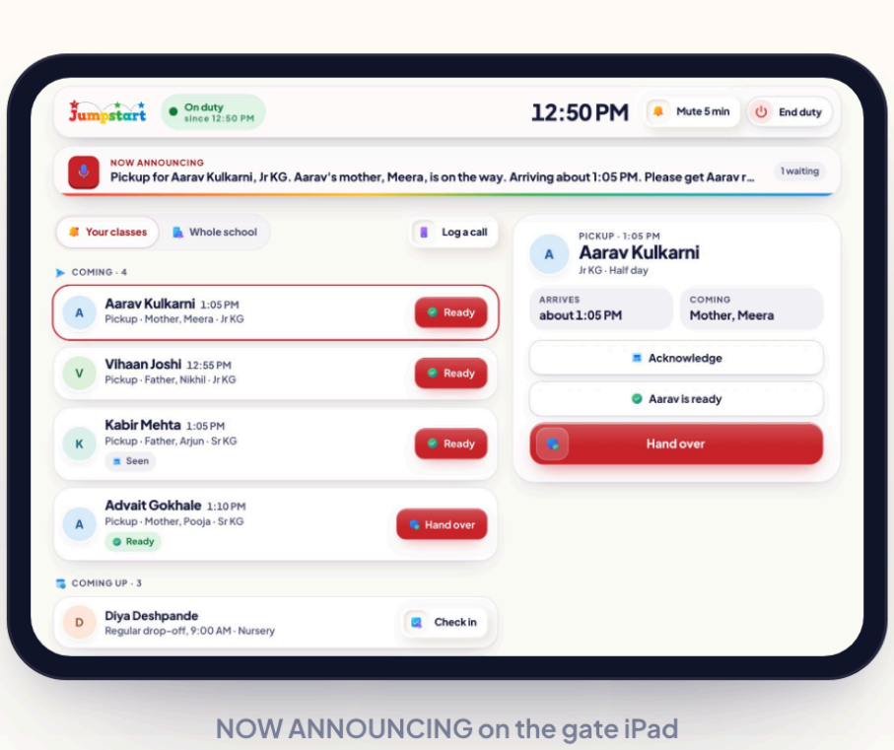
An Indian voice, made *ahead of time.*

The model runs on the school's own computer, so names never leave the school and each announcement costs nothing. Because making a sentence takes about 20 seconds, the everyday sentences for every child are made in advance and the iPad plays them instantly.

When a trip arrives at the gate

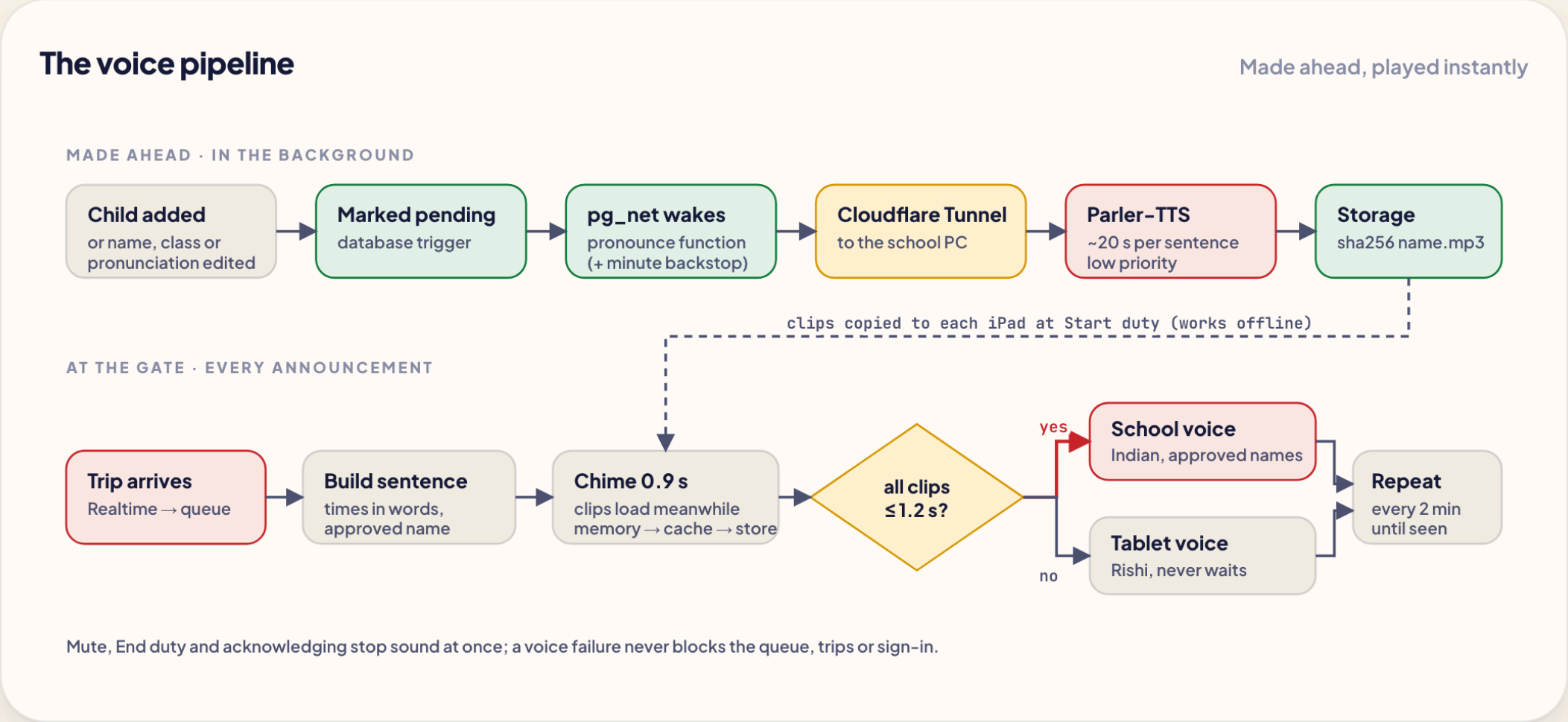
One sentence builder (`_shared/announce.ts`) is used by the gate, the admin preview and the background job, so all three produce the same words and the same clip names.

1. **The gate builds the sentence**
   "Pickup for Aarav Kulkarni, Junior K G…": times said to the nearest 5 minutes, numbers and class names written out, the child's approved pronunciation applied.
2. **Each sentence becomes a clip name**
   `sha256(voice version + words)`, so paths never contain a name and a clip never needs remaking.
3. **The chime plays (0.9 s) while clips load**
   From the fastest place that has them, below.
4. **Ready within 1.2 s? The school voice speaks**
   Otherwise the tablet's own Indian English voice (Rishi) says it, so the gate never waits. Mute, End duty and acknowledging stop it at once.
5. **Repeats every 2 minutes until acknowledged**
   "This is a reminder…", most urgent first.



NOW ANNOUNCING on the gate iPad

- `~0 ms` **1 · Memory** Decoded this session (last 120 sentences)
- `~10 ms` **2 · iPad cache** Every child's clips copied at Start duty; works offline
- `~300 ms` **3 · Supabase Storage** Private `voice-audio` bucket, staff only
- `~20 s` **4 · pronounce function** Asks the school computer for a brand-new sentence
- `fallback` **5 · Tablet voice** Rishi (Indian English) when nothing is ready in time


The voice pipeline
Made ahead, played instantly
MADE AHEAD · IN THE BACKGROUND
Child added
or name, class or pronunciation edited
Marked pending
database trigger
pg_net wakes
pronounce function (+ minute backstop)
Cloudflare Tunnel
to the school PC
Parler-TTS
~20 s per sentence low priority
Storage
sha256 name.mp3
clips copied to each iPad at Start duty (works offline)
AT THE GATE · EVERY ANNOUNCEMENT
Trip arrives
Realtime → queue
Build sentence
times in words, approved name
Chime 0.9 s
clips load meanwhile memory → cache → store
all clips ≤1.2 s?
yes
no
School voice
Indian, approved names
Tablet voice
Rishi, never waits
Repeat
every 2 min until seen

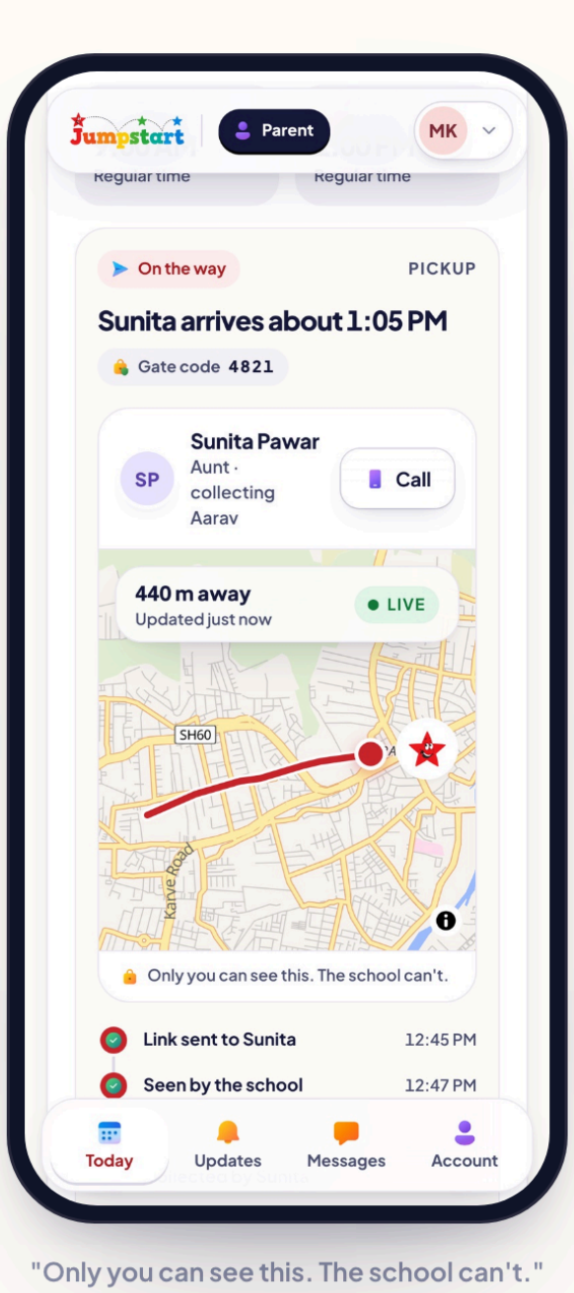
Mute, End duty and acknowledging stop sound at once; a voice failure never blocks the queue, trips or sign-in.

Made in advance: adding a child or changing a name, class, collector or pronunciation marks them "pending". `pg_net` wakes the `pronounce` function at once (and the `voice-drain` job every minute as a backstop). It asks the school computer through the Cloudflare Tunnel, saves each MP3 to Storage, and marks the child ready. The school computer runs the model at low priority so it never slows the machine.

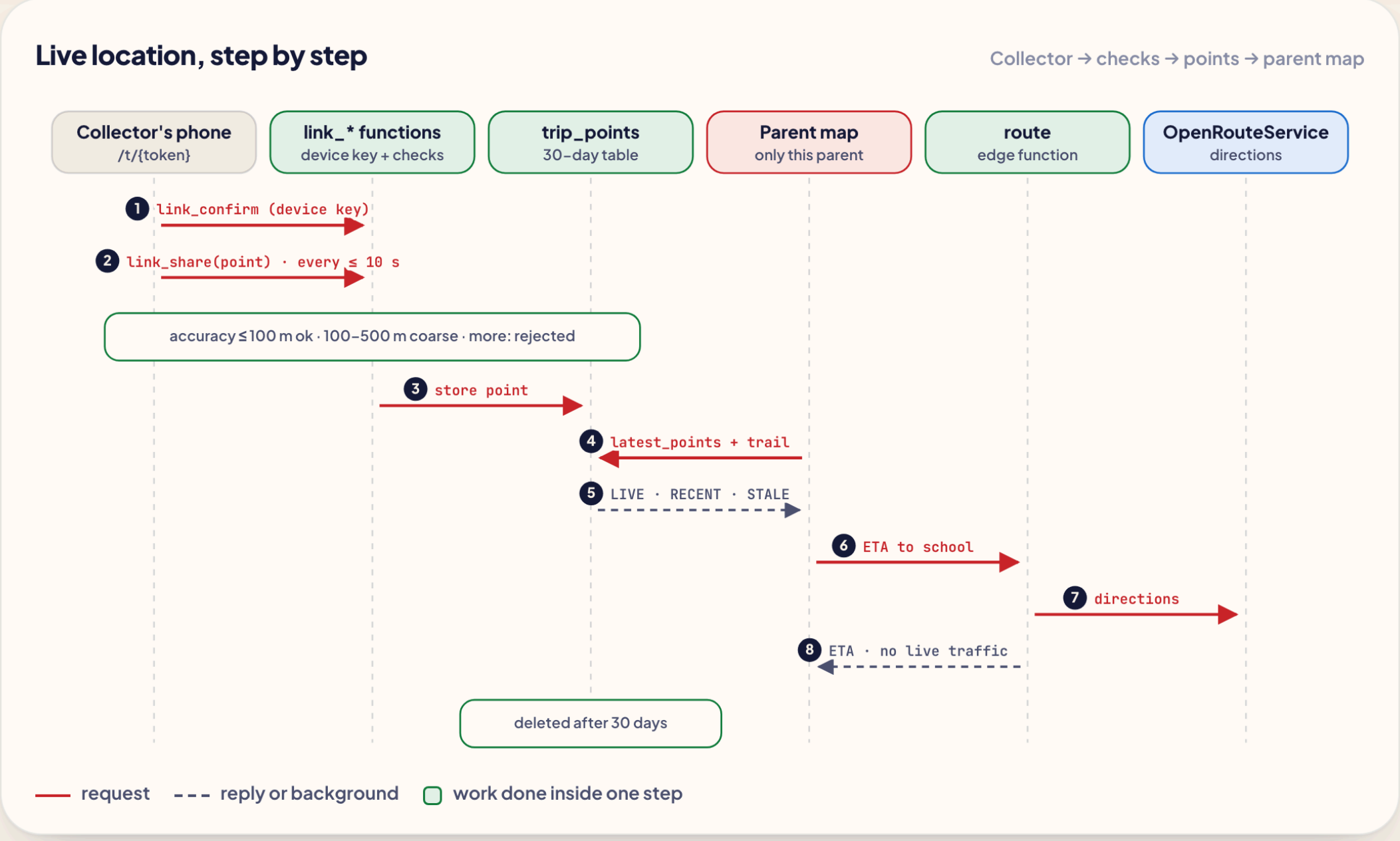
05 · SOMEONE ELSE COLLECTING

Watch them arrive, *live.*

1. **The parent sends a one-time link**
   `invite_collector` creates a token and a gate code; the parent shares it by WhatsApp or SMS from their own phone.
   Parent
2. **The collector opens `/t/{token}`**
   A small separate page, no account. `link_confirm` ties the link to that phone with a device key.
   Collector
3. **They choose to share location**
   A point when they move 10 m or more (at most every 5 s) and at least every 10 s while still. The server rejects bad accuracy; 100–500 m is stored as "coarse".
4. **The parent's map shows them**
   `latest_points` and the trail; freshness is LIVE, RECENT or STALE, and a stale marker stops moving. ETA comes from the `route` function (OpenRouteService), labelled "no live traffic".
   Only this parent
5. **Handover with the code, then the way home**
   Tracking can continue until they're home, at most an hour. Points are deleted after 30 days by `purge-trip-points`.



"Only you can see this. The school can't."



06 · THE API

Every call, *grouped.*

The app calls database functions through Supabase's API (`supabase.rpc`). Each one checks the caller's role and validates input before writing.

AREA	CALLS	WHO
Sign-in	`login_allowed` `whoami` `accept_consent`	Everyone
Trips	`start_trip` `cancel_trip` `change_day_time` `report_absence` `report_absence_range` `undo_absence`	Parents, office
Gate	`gate_step` `gate_handover_code` `gate_walk_in` `log_call` `set_attendance`	Center leaders, front desk
Collector link	`invite_collector` `collector_code` `link_open` `link_confirm` `link_share` `link_status` `link_late` `link_home` `latest_points`	Parent, collector
People	`admin_save_child` `import_child` `admin_add_person` `admin_update_person` `admin_add_guardian` `admin_child_left` `admin_add_class` `set_coordinator` `parent_add_relative`	Admin, head office
Timings	`save_programme` `set_regular_times`	Head office
Messages	`send_announcement` `send_message` `mark_read` `recipient_count`	Office, parents
Voice & push	`set_child_pronunciation` `retry_child_voice` `save_push_subscription` `set_my_notifications`	Admin, everyone

EDGE FUNCTION	WHAT IT DOES	TALKS TO
`pronounce`	Makes and stores voice clips; background drain; the voice computer registers its tunnel address here	School computer via Cloudflare Tunnel
`push-send`	Sends web push for trip events and messages	Apple, Google, Mozilla push services
`invite-send`	Invitation emails	Brevo API
`route`	Route and ETA to the school, Pune area only	OpenRouteService

TIMED JOB (PG_CRON)	PURPOSE
`morning-reminders`	Daily reminders to families
`push-flush` `push-cleanup`	Send queued push messages; drop dead subscriptions
`voice-drain`	Backstop for making pending voice clips
`purge-trip-points` `purge-old-records`	Delete location after 30 days; retention for older records

07 · SECURITY AND PRIVACY

Safe by *design.*

Rules live in the database

Row-level security on every table; a modified app can't skip a check.

Invite-only

Only people the school adds can sign in; codes expire in 5 minutes.

Location stays private

Only the parent who sent the link sees it; a child's location is never tracked.

Secrets stay server-side

Keys live only in edge-function secrets, never in the browser.

Audit log

Sensitive actions are recorded and kept for a year.

Short retention

Trips and routes 30 days; attendance this and last academic year.

Names stay in school

Announcements are made on the school's own computer; no outside voice service hears them.

Load limits

Hourly caps on new voice sentences and sign-in codes; the voice service refuses floods.

Jumpstart Go · Jumpstart Preschool, Karve Road, Pune

Screens: the real app with fictional family names · 28 September 2026